Circles - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

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Question 1 *Practice question written for this worksheet (not from a past paper)*

A circle has area 50 cm^2.

Work out its radius, to 3 significant figures.

$\pi r^2 = 50$, $r^2 = 15.915...$

$r = \sqrt{15.915}$

Answer: 3.99 cm

Question 2 *Practice question written for this worksheet (not from a past paper)*

A circle has circumference 30 cm.

Work out its radius, to 3 significant figures.

$2\pi r = 30$

$r = \frac{30}{2\pi} = 4.774...$

Answer: 4.77 cm

Question 3 *Practice question written for this worksheet (not from a past paper)*

The radius of a circle is multiplied by 3.

By what number is the area multiplied?

Area scale factor $= 3^2$

Answer: 9

Question 4 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2020 · Question 11*

Q	Working	Answer	Mark	Notes
	$2 \times \pi \times 18$ or $\pi \times 36$		2	M1
		113		A1 for 113 – 113.15
				Total 2 marks

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Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2025 · Question 13

Q	Working	Answer	Mark	Notes
	eg $\pi \times 6.4^2 \left(= \frac{1024}{25}\pi\right)$		2	M1 allow 3.14… or $\frac{22}{7}$ for π
	Correct answer scores full marks (unless from obvious incorrect working)	129		A1 accept 128 – 129
				Total 2 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 2

2.	$\pi \times 7.6$		2	M1 or $2 \times \pi \times \frac{7.6}{2}$
		23.9		A1 for answer which rounds to 23.9
				Total 2 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 16

Q	Working	Answer	Mark	Notes
	eg $\pi \times 6.3 \left(= \frac{63}{10}\pi\right)$ oe eg $2 \times \pi \times (6.3 \div 2)$ or $2 \times \pi \times 3.15$		2	M1 allow 3.14... or $\frac{22}{7}$ for π
	Correct answer scores full marks (unless from obvious incorrect working)	19.8		A1 accept 19.78 – 19.8
				Total 2 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2022 · Question 11

Q	Working	Answer	Mark	Notes
	$\frac{110}{360} \times \pi \times 7.1^2$ oe or $\frac{110}{360} \times 3.14.... \times 7.1^2$ oe		2	M1 for a complete method to find the area
		48.4		A1 accept 48.3 – 49.2
				Total 2 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 11

Q	Working	Answer	Mark	Notes
	eg $\pi \times 16^2 \times \frac{140}{360}$ oe eg $256\pi \times \frac{7}{18}$		2	M1 allow use of 3.14… or $\frac{22}{7}$ for π $\frac{140}{360}$ may be seen as an equivalent fraction or decimal eg $\frac{7}{18}$ or $0.3\dot{8}$ 0.388(8…) or 0.389
	Correct answer scores full marks (unless from obvious incorrect working)	313		A1 accept 311.8 – 313
				Total 2 marks

Circles - Mark scheme

Question 10 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 12*

Q	Working	Answer	Mark	Notes
	$\frac{50}{360}\times\pi\times7\times2$ oe eg $\frac{14\pi}{36}\times5$ or "43.98…" ÷ 360 × 50 oe		2	M1 Students may use π or 3.14, 3.142 or $\frac{22}{7}$
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	6.1		A1 Accept answers in the range 6.05 – 6.2
				Total 2 marks

Question 11 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 15*

15	$\frac{75}{360}\times\pi\times12^2(=30\pi)$ oe		2	M1 a correct method to find the area of the sector Allow 3.14… or $\frac{22}{7}$ for π
	Correct answer scores full marks (unless from obvious incorrect working)	94.2		A1 accept 94.2 – 94.3
				Total 2 marks

Question 12 *Practice question written for this worksheet (not from a past paper)*

A circle of radius 5 cm fits exactly inside a square of side 10 cm.

Work out the area of the square not covered by the circle. Give your answer to 3 s

Square: $10 \times 10 = 100$

Circle: $\pi \times 5^2 = 78.54$

$100 - 78.54$

Answer: 21.5 cm²

Question 13 *Practice question written for this worksheet (not from a past paper)*

A rectangle is 12 cm by 8 cm. A circle of diameter 6 cm is cut out of it.

Work out the area that is left, to 3 significant figures.

Rectangle: 96

Circle: $\pi \times 3^2 = 28.27$

$96 - 28.27$

Answer: 67.7 cm²

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Question 14 *Practice question written for this worksheet (not from a past paper)*

A circular pond has radius 4 m. A path 1 m wide goes all the way round it.

Work out the area of the path, to 3 significant figures.

Outer radius 5 m: $\pi \times 5^2 = 78.54$

Pond: $\pi \times 4^2 = 50.27$

$78.54 - 50.27$

Answer: 28.3 m²

Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 5*

5	$\pi \times 20 (= 20\pi = 62.8(31...))$ oe or $2 \times \pi \times (20 \div 2) (= 20\pi = 62.8(31...))$ oe or $0.5 \times \pi \times 20 (= 10\pi = 31.4(15...))$ oe or		3	M1 for use of $\frac{1}{2}\pi d$ or πr or πd or $2\pi r$ oe with $d = 20$ or $r = 20 \div 2$ (= 10)
	3 × ("62.8" ÷ 2) + 20 ÷ 2 + 20 ÷ 2 oe or 1.5 × ("62.8") + 20 ÷ 2 + 20 ÷ 2 oe			M1 for a complete method
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	114		A1 114 – 115 SCB1 for awrt 471
				Total 3 marks

Question 16 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 13*

13	eg $2 \times \pi \times 5.2 (= 32.6...$ or $\frac{52}{5}\pi)$ oe $\frac{67}{360} \times 2 \times \pi \times 5.2 (= 6.08...$ or $\frac{871}{450}\pi)$ oe		3	M1	for finding the whole circumference or the arc length
	$\frac{67}{360} \times 2 \times \pi \times 5.2 + 2 \times 5.2$ oe			M1	for a complete method
		16.5		A1	accept 16.4 - 16.5 (not in terms of π)
					Total 3 marks

Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 13*

13	$\frac{60}{360} \times 2 \times \pi \times r$ oe or $\frac{1}{6} \times 2 \times \pi \times r$ oe		3	M1	for finding the length of the arc
	"$\frac{60}{360} \times 2 \times \pi \times r$"$+ 2r$ oe or "$\frac{1}{6} \times 2 \times \pi \times r$"$+ 2r$ oe			M1	dep on M1 for a complete expression from correct working for a method for the perimeter
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$P = r\left(\frac{1}{3}\pi + 2\right)$		A1	oe eg $P = r(0.33...\pi + 2)$ or $P = \left(\frac{1}{3}\pi + 2\right)r$ or $P = \left(2 + \frac{120}{360}\pi\right)r$ or $P = \left(\frac{120}{360}\pi + 2\right)r$
					Total 3 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 14

14	$(\angle AOC =) 132 \times 2 (= 264)$		3	M1	for method to find angle at the centre. Do not award this mark if contradicted on the diagram eg if obtuse AOC is labelled as 264
	eg $\frac{"264"}{360} \times 2 \times \pi \times 8.5$ (= 39.1... or $\frac{187}{15}\pi$) **or** $2 \times \pi \times 8.5 - \frac{360 - "264"}{360} \times 2 \times \pi \times 8.5$ (= 39.1... or $\frac{187}{15}\pi$) **or** $\frac{"264"}{360} \times 2 \times \pi \times 8.5 + 2 \times 8.5$ **or** $2 \times \pi \times 8.5 - \frac{360 - "264"}{360} \times 2 \times \pi \times 8.5 + 2 \times 8.5$			M1	for a method to find the length of arc AC or perimeter of the sector – allow use of their AOC as long as clearly labelled
	Correct answer scores full marks (unless from obvious incorrect working)	56.2		A1	accept 56.1 – 56.2
					Total 3 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2018 · Question 6

6	$9.7^2 + 3.5^2 (=106.34)$	32.4	4	M1	M1 for the use of MN and a correct angle (70.1… or 70.2, 19.8…) in a correct trig statement eg $\cos 70.2 = \frac{3.5}{MN}$
	$\sqrt{9.7^2 + 3.5^2}$ or $\sqrt{"106.34"}$ (=10.3…)			M1	M1 for a complete method to find MN eg $MN = \frac{3.5}{\cos 70.2}$ (=10.3…)
	$\pi \times$ "10.3…" **or** $2 \times \pi \times \frac{"10.3..."}{2}$			M1	dep on M2
				A1	for answer in range 32.3 – 32.41

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2021 · Question 11

11	$360 - 40 (= 320)$ or $\frac{320}{360}$ oe **or** $\frac{40}{360} \times 2\pi \times 9$ (= 6.28...)		4	M1	
	$\frac{"320"}{360} \times 2\pi \times 9 (= 16\pi = 50.26...)$ **or** $2\pi \times 9$ – "6.28" (= 50.26)			M1	
	"50.26" + 2 × 9			M1	complete method to find perimeter
		68.3		A1	68.2 to 68.3
					Total 4 marks

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Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2023 · Question 25

Q	Working	Answer	Mark	Notes
	$\sin 52 = \frac{12 \div 2}{r}$ oe or $\frac{r}{\sin 90} = \frac{6}{\sin 52}$ oe or $\cos(90-52) = \frac{12 \div 2}{r}$ oe or $(r^2 =)(12 \div 2)^2 + \left(\frac{12 \div 2}{\tan 52}\right)^2$ oe $\left[r^2 = 6^2 + 4.687...^2\right]$ or $\frac{r}{\sin 38} = \frac{12}{\sin 104}$ oe		4	M1 A correct trig statement for the radius use of tan must also include a correct Pythagoras statement.
	$r = \frac{6}{\sin 52}$ (=7.614) oe or $r = \frac{6}{\cos 38}$ oe or $(r =)\sqrt{(12 \div 2)^2 + \left(\frac{12 \div 2}{\tan 52}\right)^2}$ $\left[r = \sqrt{6^2 + 4.687...^2}\right]$ oe or $\frac{12 \sin 38}{\sin 104}$ oe			M1 A correct method to find the radius of the circle use of tan must also use Pythagoras to find an expression for r
	(Area =) $\pi \times ("7.61...")^2$			M1 the radius must come from a completely correct method
	Correct answer scores full marks (unless from obvious incorrect working)	182		A1 Accept 181 - 183
				Total 4 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 15

15	$\frac{80}{360} \times \pi \times r^2 = 32\pi$ oe		4	M1 for forming a correct equation, allow use of any variable NB: treat consistent use of 32 instead of 32π as the area as a misread (all M marks available)
	$(r =)\sqrt{\frac{32\pi \times 360}{80 \times \pi}}$ $(=\sqrt{144} = 12)$			M1 for a correct method to solve for r NB: misread of the area as 32 gives $r = 6.7(7\ldots)$
	$\frac{80}{360} \times 2 \times \pi \times "12"$ $\left(= \frac{16}{3}\pi = 16.7(551...)\right)$ oe			M1 substitution of r from a correct method to find the arc length NB: misread of the area as 32 gives arc length = awrt 9.5
	Correct answer scores full marks (unless from obvious incorrect working).	40.8		A1 allow an answer in the range 40.7 – 40.8
				Total 4 marks

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Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 16

Q	Working	Answer	Mark	Notes
	$120 = \frac{1}{2} \times a \times b \times \sin 50$ oe or $\frac{120 \times 2}{\sin 50}$ oe or $120 = \frac{1}{2} \times 2x \sin 25 \times x \cos 25$ oe or $\frac{120 \times 2}{2 \times \sin 25 \times \cos 25}$ oe or 313(.2977494)		4	M1 for a correct equation for $a \times b$ or r^2 (allow any letters for a or b) or for a correct expression for $a \times b$ or r^2 or for 313(.2977494)
	$(radius =) \sqrt{\frac{120 \times 2}{\sin 50}}$ (= 17.7(0021891)) oe or $(radius =) \sqrt{\frac{120 \times 2}{2 \times \sin 25 \times \cos 25}}$ (= 17.7(0021891)) oe or $(radius =) \sqrt{313(.2977494)}$ (= 17.7(0021891))			M1 for a correct rearrangement to find the radius or for square rooting 313(.2977494) or for 17.7(0021891))
	(area of $OAPB$ =) $\pi \times$ "17.7"$^2 \times \frac{50}{360}$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	137		A1 awrt 137
				Total 4 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2022 · Question 16

16	eg $\frac{55}{360} \times \pi \times d = 5$ **or** $\frac{55}{360} \times \pi \times 2 \times r = 5$ oe **OR** $\frac{360}{55} \times 5 (= 32.7...)$ oe		4	M1 for a correct equation for the diameter **or** radius **OR** for a method to find the circumference of the circle
	eg $d = \frac{5 \times 360}{55\pi} (= 10.4...)$ **or** $r = \frac{5 \times 360}{55 \times 2 \times \pi} (= 5.2...)$ **OR** $d = \frac{32.7...}{\pi} (= 10.4...)$ **or** $r = \frac{32.7...}{2 \times \pi} (= 5.2...)$			M1 for a method to work out the diameter **or** radius
	(area =) eg $\pi \times \left(\frac{"10.4..."}{2}\right)^2$ **or** $\pi \times$ "5.2..."^{2}			M1
		85.2		A1 allow 84.9 – 85.4
				Total 4 marks

Question 25

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 19

19	$\pi \times 12^2 \times \frac{AOC}{360} (= 100)$		4	M1 oe for setting up a correct expression for the area of the sector (or equation)
	$(AOC =) \frac{100 \times 360}{\pi \times 12^2} \left(= \frac{250}{\pi}\right)$			M1 for correctly rearranging for AOC
	(Angle ABC =) "79.57747" ÷ 2 (= 39.7887… or $\frac{125}{\pi}$)			M1 ft dep 1st M1 **and** 'x' less than 360 for dividing their 'AOC' by 2
		39.8		A1 for awrt 39.8 accept $\frac{125}{\pi}$
				Total 4 marks

Question 26

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 20

Q	Working	Answer	Mark	Notes
	eg $\pi r^2 \times \frac{60}{360} - \frac{1}{2} r^2 \sin 60$ oe or $\frac{\pi r^2}{6} - \frac{\sqrt{3}}{4} r^2$ oe		4	M1 for a correct expression for the area of the segment Expression may be embedded in an equation, eg $\pi r^2 \times \frac{60}{360} - \frac{1}{2} r^2 \sin 60 = 38$ or $\pi r^2 \times \frac{60}{360} = 38 + \frac{1}{2} r^2 \sin 60$ or $\pi r^2 \times \frac{60}{360} - 38 = \frac{1}{2} r^2 \sin 60$
	eg $(r^2 =) 38 \div \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right) (= 38 \div 0.09(058))$ (= 419(.490…)) oe or $(r =) \sqrt{38 \div \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)}$ (= 20.4(81…))oe			M1 dep on M1 for a correct expression for r^2 or r
	$\frac{\pi}{6} \times$"20.4(81...)"$\times 2$ (= 21.4(48…)) oe or			M1 for using the value of r to find arc length
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	41.9		A1 allow 41 - 42
				Total 4 marks

Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2017 · Question 22

22	$\frac{75}{360} \times \pi \times d\,(2r) = 7.2$		4	M1	for a correct equation linking the angle and arc length NB: 0.208(3…) may be used in place of $\frac{75}{360}$ **or** 4.8 in place of $\frac{360}{75}$
	$\frac{7.2 \times 360}{75 \times \pi}$ oe **or** $\frac{7.2 \times 360}{75 \times 2 \times \pi}$ oe **or** $d = 11(.0…)$ **or** $r = 5.5(0…)$			M1	for a complete method to find the radius or diameter.
	$\frac{75}{360} \times \pi \times \left(\frac{"11"}{2}\right)^2$ **or** $\frac{75}{360} \times \pi \times "5.5"^2$ **or** $\frac{75}{360} \times 95(.04...)$			M1	dep on previous M1
		19.8		A1	for answer in range 19.8 – 19.82
					Total 4 marks

CHALLENGE QUESTIONS

Question 28 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 20*

	eg $(AB^2 =) 6^2 + 6^2 (= 72)$ or $\sin 45 = \frac{6}{(AB)}$ or $\cos 45 = \frac{6}{(AB)}$ or or $(AB^2 =) 6^2 + 6^2 - 2\times 6\times 6\times \cos 90$		5	M1	for a correct start to the method to find AB
	eg $(AB =)\sqrt{6^2+6^2}\ (=\sqrt{72}$ or $6\sqrt{2}$ or $8.48...)$ or $(AB =)\frac{6}{\sin 45}\ (=\sqrt{72} = 6\sqrt{2} = 8.48...)$ or $(AB =)\frac{6}{\cos 45}\ (=\sqrt{72} = 6\sqrt{2} = 8.48...)$ or $(AB =)\sqrt{6^2+6^2-2\times 6\times 6\times \cos 90}$			M1	for a complete method to find the length of AB
	eg $\pi\times 6 (= 6\pi$ or $18.8...)$ or $\pi\times 6\div 2 (= 3\pi$ or $9.42...)$ or $\pi\times$"8.48..."$(= 26.6...)$ or $\pi\times$"8.48..."$\div 2\ (=13.3...)$			M1	(indep) for a method to find the circumference of one whole circle **or** the arc length of one semicircle seen (may be embedded)
	eg $2\times$"3π"+"13.3..." **or** "9.42" + "9.42" + "13.3" **or** "18.8" + "13.3"			M1	for a complete correct method to find the perimeter of the shape
	Correct answer scores full marks (unless from obvious incorrect working)	32.2		A1	accept answers in the range 32.1 – 32.3
					Total 5 marks

Question 29 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 19*

19	$\pi\times(r+7)^2\times\frac{45}{360}$ oe or $(2\times)\pi\times(r-2)^2$ oe		5	M1
	$\pi\times(r+7)^2\times\frac{45}{360} = 2\times\pi\times(r-2)^2$ oe			M1 for a correct equation
	E.g. $675r^2 - 3510r + 675 (= 0)$ $15r^2 - 78r + 15 (= 0)$ oe or $5r^2 - 26r + 5 (= 0)$ oe Allow $5r^2 - 26r = -5$ or $[4(r-2)]^2 = (r+7)^2$ or $(r-2)^2 = \left[\frac{(r+7)}{4}\right]^2$			A1 (dep on M2) writing a correct quadratic expression in form $ax^2 + bx + c\ (= 0)$ allow $ax^2 + bx = c$
	$(5r-1)(r-5)(=0)$ oe or $(r =)\ \frac{--26\pm\sqrt{(-26)^2-4\times 5\times 5}}{2\times 5}$ or $5\left(\left(r-\frac{26}{10}\right)^2-\left(\frac{26}{10}\right)^2\right)+5=0$ oe or $4r-8 = r+7$ oe			M1 (dep on M1) for a complete method to solve their 3-term quadratic equation Allow one sign error and some simplification – allow as far as $\frac{26+\sqrt{676-100}}{10}$
		5		A1 dep on M2 (5 and $\frac{1}{5}$ scores M1M1A1M1A0)
				Total 5 marks

Question 30

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2021 · Question 22

22		eg $(AD=)\sqrt{6^2+6^2-2\times6\times6\times\cos(50)}$ $(=5.07\ldots)$**or** $2\times6\sin25$ (=5.07...) **or** $\frac{6\sin50}{\sin65}(=5.07\ldots)$ oe		6	M1	Correct expression for AD ie $AD=\ldots$ or $x=$ oe
		eg $6+6+\sqrt{6^2+6^2-2\times6\times6\times\cos(50)}$ or $12+$"5.07…" (=17.0)7... or 17.1)			M1	A **correct** statement of perimeter of triangle OAD
		eg (arc $BC=$) $\frac{50}{360}\times\pi\times2\times(6+x)$ oe			M1	A correct statement for arc BC (condone missing brackets around $(6+x)$ for this mark only)
		eg $2\times"17.1"=12+2x+\frac{50}{360}\times\pi\times2\times(6+x)$ oe			M1	dep on M3 for a correct equation for x
		eg $2\times17.1-12-\frac{30}{18}\pi=2x+\frac{5x}{18}\pi$			M1	isolating terms in x in a correct equation
			5.89		A1	5.88 – 5.89
						Total 6 marks

Question 31

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 25

Q	Working	Answer	Mark	Notes
	[chord AB =]$\sqrt{5^2+5^2-2\times5\times5\times\cos50}$ **or** $2\times5\times\sin25$ (= 10sin25 or 4.226...)		6	M1 oe
	[$\angle APB$ =]$\cos^{-1}(\frac{4^2+4^2-"4.226..."^2}{2\times4\times4})$ (=63.77...) **or** [$\angle OPA$ =]$\sin^{-1}(\frac{0.5\times"4.226..."}{4})$ (= 31.88...)			M1 oe may use other methods but must be a complete method for $\angle APB$ **or** $\angle OPA$ (see below for sine rule)
	[Area sector AOB =] $\frac{50}{360}\times\pi\times5^2(=\frac{125}{36}\pi$ or 10.9...)			M1 oe independent
	[Area sector APB =]$\frac{"63.77..."}{360}\times\pi\times4^2$ (=8.90...)			M1 oe NB: $2\times$ "31.88…" = "63.77…"
	$\left(\frac{50}{360}\pi\times5^2-\frac{1}{2}\times5^2\times\sin50\right)+\left(\frac{"63.77..."}{360}\times\pi\times4^2-\frac{1}{2}\times4^2\times\sin"63.77..."\right)$			M1 oe (10.9…– 9.57…) + (8.90… – 7.17…)
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	3.06		A1 allow 3 – 3.1
	Alternative version (using line of symmetry OP in quadrilateral $OAPB$)			**Total 6 marks**
	[$\angle OPA$] $=\sin^{-1}\left(\frac{5\sin25}{4}\right)$ (= 31.88..)		6	M1 oe (see above for cosine rule & trig)
	[Area sector APB =] $\frac{2\times"31.88..."}{360}\times\pi\times4^2$ (= 8.90...)			M1 oe
	[Area $OAPB$ =] 2 × ½ × 5 × 4 × sin(180 – "31.88.." – 25) (=16.75...)			M1 oe
	[Area sector AOB =] $\frac{50}{360}\times\pi\times5^2(=\frac{125}{36}\pi=10.9\ldots)$			M1 oe independent
	[Area **R** =] "10.9..." + "8.90..." – "16.75..."			M1 oe
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	3.06		A1 allow 3 – 3.1
				Total 6 marks

Question 32

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 25

25	eg $\frac{40}{360}\pi r^2 - \frac{1}{2}r^2 \sin 40 (= 28)$ oe **or** $\frac{40}{360}\pi r^2 = 28 + \frac{1}{2}r^2 \sin 40$ oe		6	M1	for a correct expression for the area of the shaded region Allow 3.14… or $\frac{22}{7}$ for π sin 40 = 0.64…
	(radius ^{2} =) 992 – 1024 (radius =) 31.8(096…)	31.8		A1	Allow answers in the range 31.5 – 32.0
	eg $(BC^2 =) 2\times"31.8"^2 - 2\times"31.8"^2 \cos 40 (= 473.4...)$ **or** $\frac{0.5BC}{"31.8"} = \sin 20$ **or** $\frac{BC}{\sin 40} = \frac{"31.8"}{\sin(70)}$			M1	for a correct first step to find *BC* using **their clearly identified radius** eg *r* = …. or seen on diagram NB $\frac{180-40}{2} = 70$ sin 20 = 0.34… sin 70 = 0.93… or 0.94
	eg $(BC =)\sqrt{2\times"31.8"^2 - 2\times"31.8"^2 \cos 40} (= 21.7...)$ **or** $(BC =) 2\times"31.8"\sin 20 (= 21.7...)$ **or** $BC = \frac{"31.8"\sin 40}{\sin(70)} (= 21.7...)$			M1	dep on previous M1 for a complete method to find *BC* cos 40 = 0.76… or 0.77
	eg $\frac{130}{360}\times 2\times\pi\times"21.7"$			M1	dep on previous M1 for a complete method to find the length of arc *BD*
	Correct answer scores full marks (unless from obvious incorrect working)	49.4		A1	accept 48.9 – 49.7
					Total 6 marks

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